For every billion annihilations—leaving a billion photons in their wake—a single hadron survived. Those loners would ultimately get to have all the fun: serving as the ultimate source of matter to create galaxies, stars, planets, and petunias.

Without the billion-and-one to a billion imbalance between matter and antimatter, all mass in the universe would have self-annihilated, leaving a cosmos made of photons *and nothing else*—the ultimate let-there-be-light scenario.

✴

By now, one second of time has passed.

✴

The universe has grown to a few light-years across,[††] about the distance from the Sun to its closest neighboring stars. At a billion degrees, it's still plenty hot—and still able to cook electrons, which, along with their positron counterparts, continue to pop in and out of existence. But in the ever-expanding, ever-cooling universe, their days (seconds, really) are numbered. What was true for quarks, and true for hadrons, had become true for electrons: eventually only one electron in a billion survives. The rest annihilate with positrons, their antimatter sidekicks, in a sea of photons.

Right about now, one electron for every proton has been "frozen" into existence. As the cosmos continues to cool—dropping below a hundred million degrees—protons fuse with protons as well as with neutrons, forming atomic nuclei and hatching a universe in which ninety percent of these nuclei are hydrogen and ten percent are helium, along with trace amounts of deuterium ("heavy" hydrogen), tritium (even heavier hydrogen), and lithium.

✴

Two minutes have now passed since the beginning.

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For another 380,000 years not much will happen to our particle soup. Throughout these millennia the temperature remains hot enough for